Clinical Microbiology and Infection 30 (2024) 328–335

Contents lists available at ScienceDirect

Clinical Microbiology and Infection

journal homepage: www.clinicalmicrobiologyandinfection.com

Systematic review

Prevalence and risk factors for persistent symptoms after COVID-19: a systematic review and meta-analysis

De Luo [1, 2], Bingjie Mei [3], Piao Wang [1, 2], Xujia Li [1, 2], Xinpei Chen [4], Gang Wei [5], Fei Kuang [6], Bo Li [1, 2], Song Su [1, 2, *]

[1)] *Department of General Surgery (Hepatopancreatobiliary Surgery), Affiliated Hospital of Southwest Medical University, Sichuan, China*
[2)] *Academician (Expert) Workstation of Sichuan Province, Metabolic Hepatobiliary and Pancreatic Diseases Key Laboratory of Luzhou City, Affiliated Hospital of Southwest Medical University, Sichuan, China*
[3)] *School of Medicine, University of Electronic Science and Technology of China, Sichuan Cancer Hospital, Sichuan, China*
[4)] *Department of Hepatobiliary Surgery, People's Hospital of Deyang City, Deyang, China*
[5)] *Department of Cardiology, Affiliated Hospital of Southwest Medical University, Sichuan, China*
[6)] *Key Laboratory of Carcinogenesis and Translational Research, Gastrointestinal Cancer, Peking University Cancer Hospital and Institute, Beijing, China*

ARTICLE INFO

Article history:
Received 13 June 2023
Received in revised form
22 September 2023
Accepted 16 October 2023
Available online 20 October 2023

Editor: M. Paul

Keywords:
COVID-19
Long COVID
Post-COVID-19 sequelae
Post-COVID-19 symptoms
Prevalence
Risk factors
Systematic review

ABSTRACT

Background: Long-term physical and mental persistent symptoms after COVID-19 represent a growing global public health concern. However, there remains a substantial knowledge gap regarding their prevalence and risk factors.
Objectives: To estimate the prevalence and risk factors for persistent symptoms after COVID-19.
Methods of data synthesis: We used a random-effects model to pool persistent symptom prevalence and risk ratios comparing COVID-19 patients with non-COVID-19 individuals.
Data Sources: Electronic databases were searched for studies published from December 2019 to January 2023.
Study eligibility criteria: Eligible studies that reported the prevalence and risk factors for persistent symptoms after COVID-19 were included.
Participants: Patients who recovered from COVID-19.
Assessment of risk of bias: The Joanna Briggs Institute critical appraisal tool was used to assess the risk of bias in prevalence studies, whereas the risk of bias in cohort studies was evaluated with the Newcastle-Ottawa Scale.
Results: After screening 4359 studies, a total of 211 eligible studies were included, covering a population of 13 368 074 individuals. Fatigue, dyspnoea, post-traumatic stress disorder, anxiety, and depression were the most frequently reported persistent symptoms after COVID-19. Subgroup analyses revealed that individuals with more severe illness in the acute phase or from Europe exhibited a higher prevalence of certain symptoms, whereas children demonstrated a lower prevalence. Furthermore, COVID-19 patients had a significantly higher prevalence of most persistent symptoms compared with non-COVID-19 individuals. Factors frequently associated with a higher prevalence of persistent symptoms included female gender, advanced age, severe illness during the acute phase of COVID-19, multiple comorbidities, an extended duration of hospital stay, and a high body mass index.
Conclusion: This meta-analysis provides a thorough review of the prevalence and risk factors for persistent symptoms following COVID-19. The findings underscore the importance of long-term monitoring and support for individuals recovering from COVID-19. **De Luo, Clin Microbiol Infect 2024;30:328**
© 2023 European Society of Clinical Microbiology and Infectious Diseases. Published by Elsevier Ltd. All rights reserved.

* Corresponding author. Song Su, Department of General Surgery (Hepatopancreatobiliary surgery), Affiliated Hospital of Southwest Medical University, Sichuan 646000, China.
E-mail address: susong1978@swmu.edu.cn (S. Su).

https://doi.org/10.1016/j.cmi.2023.10.016
1198-743X/© 2023 European Society of Clinical Microbiology and Infectious Diseases. Published by Elsevier Ltd. All rights reserved.

Introduction

After acute COVID-19, many patients continue to experience lasting physical, psychological, or cognitive symptoms, often referred to as long COVID [1,2]. Long COVID is defined by the United Kingdom National Institute for Health and Care Excellence as the presence of symptoms that extend beyond 4 weeks after SARS-CoV-2 infection. It encompasses ongoing symptomatic COVID-19 persisting from 4 to 12 weeks after infection, and post-acute COVID-19 syndrome lasting beyond 12 weeks [3,4]. An estimated minimum of 65 million individuals worldwide are affected by long COVID, considering a conservative incidence rate of 10% among infected individuals and the existence of over 651 million documented COVID-19 cases globally [5]. Long COVID has been recognized as a potential public health disaster, emphasizing the urgent need to understand the prevalence and risk factors associated with persistent symptoms after COVID-19 and assisting health care providers in developing better strategies for managing these patients.

Previous meta-analyses conducted early in the pandemic have investigated the prevalence of persistent symptoms [6–9]. However, these studies exhibit several limitations:(a) They solely concentrate on specific populations, such as hospitalized or young individuals; (b) They include studies with short follow-up periods or small sample sizes; (c) They fail to compare these symptoms between COVID-19 patients and non-COVID-19 individuals; (d) They do not perform subgroup analyses based on factors that influence prevalence to explore the characteristics of long COVID within these distinct populations; and (e) They do not analyse the risk factors associated with long COVID. Considering that 3 years have elapsed since the onset of the pandemic, a multitude of large-scale, high-quality studies investigating post-COVID-19 condition, with extensive follow-up periods, have been undertaken. It is imperative to systematically and comprehensively synthesize the worldwide data on long COVID in order to acquire the most current evidence for guiding clinical practice.

The objectives of this study were as follows: (a) To investigate the prevalence of persistent symptoms following COVID-19; (b) To assess the prevalence of persistent symptoms at different follow-up periods after COVID-19; (c) To compare the incidence of persistent symptoms between individuals with a history of COVID-19 and those without; and (d) To identify risk factors associated with persistent symptoms after COVID-19. Moreover, subgroup analyses were performed considering age, geographic location, and the severity of acute COVID-19.

Methods

This systematic review and meta-analysis was registered in the international prospective register of systematic reviews (PROSPERO) [10] with the registration number CRD42022345114.

Search strategy

Three investigators (DL, PW, and JXL) independently performed a systematic search in four electronic databases (PubMed, Embase, Web of Science, and Cochrane Library) to identify eligible studies on long COVID published from December 2019 to January 2023. In addition, the references of the included studies were manually searched to identify any relevant studies. Detailed information on the search strategy can be found in Table S1.

Outcomes of interest

Our outcomes of interest were prevalence and risk factors for persistent symptoms, which are defined in Table S2.

Study selection

Three investigators (PW, XJL, and GW) independently screened the titles and abstracts of all studies, and any disagreements were resolved through consensus. The inclusion criteria for eligible studies were as follows: (a) inclusion of patients with confirmed evidence of SARS-CoV-2 infection through PCR testing, antibody testing, or clinical diagnosis; (b) a minimum follow-up period of 4 weeks since COVID-19 diagnosis, assuming that events such as hospital admission or discharge, resolution of acute illness, or symptom onset occurred either concurrently or subsequent to diagnosis; (c) reporting of at least one of the outcomes of interest; and (d) study design as cohort or cross-sectional. This study also included studies that compared the prevalence of symptoms between the COVID-19 group and the comparator group. The COVID-19 group included patients with laboratory-confirmed COVID-19. The comparator group included community individuals without COVID-19, health workers from hospitals, and matched hospitalized patients because of pneumonia or respiratory infection with negative COVID-19 PCR.

We excluded studies that were only available as preprints and lacked a peer-review process. In addition, studies with incomplete or imprecise quantitative data were excluded. To ensure sufficient statistical power and precision, we also excluded studies that did not meet the predetermined sample size threshold of 121, as calculated using PASS 11 software (with a confidence level of 0.95, a confidence interval width of 0.17, and an assumed proportion of 0.3). This sample size threshold aimed to ensure that the included studies had adequate power to achieve a margin of error of at least 0.05 for the provided prevalence estimate.

Data extraction

Three investigators (PW, XJL, and GW) independently conducted data extraction and documented the findings using a predefined form. The extracted information included the first author, publication year, country, study design, demographic characteristics, sample size, method of COVID-19 confirmation, index date, follow-up periods, and investigated outcomes. In instances of missing information, the authors of the corresponding studies were contacted to acquire the required details. Subsequently, two investigators cross-validated the extracted data, and any discrepancies were resolved through discussion and consensus.

Quality assessment

Two independent investigators (PW and XJL) assessed the methodological quality of the included studies using specific risk of bias assessment tools tailored to each study design. Consensus was reached to resolve any discrepancies between the investigators. The risk of bias in each prevalence study was evaluated using the Joanna Briggs Institute (JBI) critical appraisal tool [11]. The methodological quality of cohort studies was assessed using the Newcastle-Ottawa Scale (NOS) [12]. Details of the JBI critical appraisal tool and NOS are described in Tables S3 and S4.

Statistical methods

Our primary analyses aimed to determine the pooled prevalence of each persistent symptom after COVID-19. To ensure data normality, the double arcsine method was used to transform the data from the studies before pooling them in the prevalence meta-analysis. For the secondary analysis, we computed risk ratios to compare the prevalence of persistent symptoms between COVID-19 patients and non-COVID-19 individuals. Because of the variability in reported symptom patterns and numbers across studies, meta-analysis was conducted only for symptoms with data available from at least five studies. Statistical analyses were performed using Stata version 16.

To account for potential between-study heterogeneity and its effect on the analysis, we used a random-effects model in the meta-analysis. Heterogeneity across studies was assessed using the I^2 statistic, where values above 75% were considered high, values between 25% and 75% were considered moderate, and values below 25% were considered low [13]. Cochran's Q-statistics, with a significance threshold of $p < 0.1$, were also used to identify the presence of heterogeneity. To investigate heterogeneity between studies, subgroup meta-analyses were conducted based on age, geographic location, and severity of acute COVID-19. Moreover, subgroup analysis was used to assess the effect of these factors on the prevalence of persistent symptoms. For fixed-effect meta-analyses, we assessed differences between subgroups by interaction tests as described by Deeks 2001 [14]. For random-effects meta-analyses, we assessed differences between subgroups by inspection of the subgroups' confidence intervals. Non-overlapping confidence intervals for any two subgroups indicated a statistically significant difference in the prevalence of persistent symptoms between the subgroups [15–17]. The follow-up periods were divided into four intervals: <3 months, 3–6 months, 6–12 months, and ≥12 months. Separate meta-analyses were performed for each interval. Statistical significance was defined as $p < 0.05$.

Results

Characteristics of the included studies

Total of 4359 titles were identified through database searches. Ultimately, 211 studies were included in this meta-analysis (Fig. 1) after removing duplicate documents and screening abstracts and full texts. Among the included studies, 28 compared the prevalence of persistent symptoms between individuals with and without COVID-19 infection. The references included in this study are listed

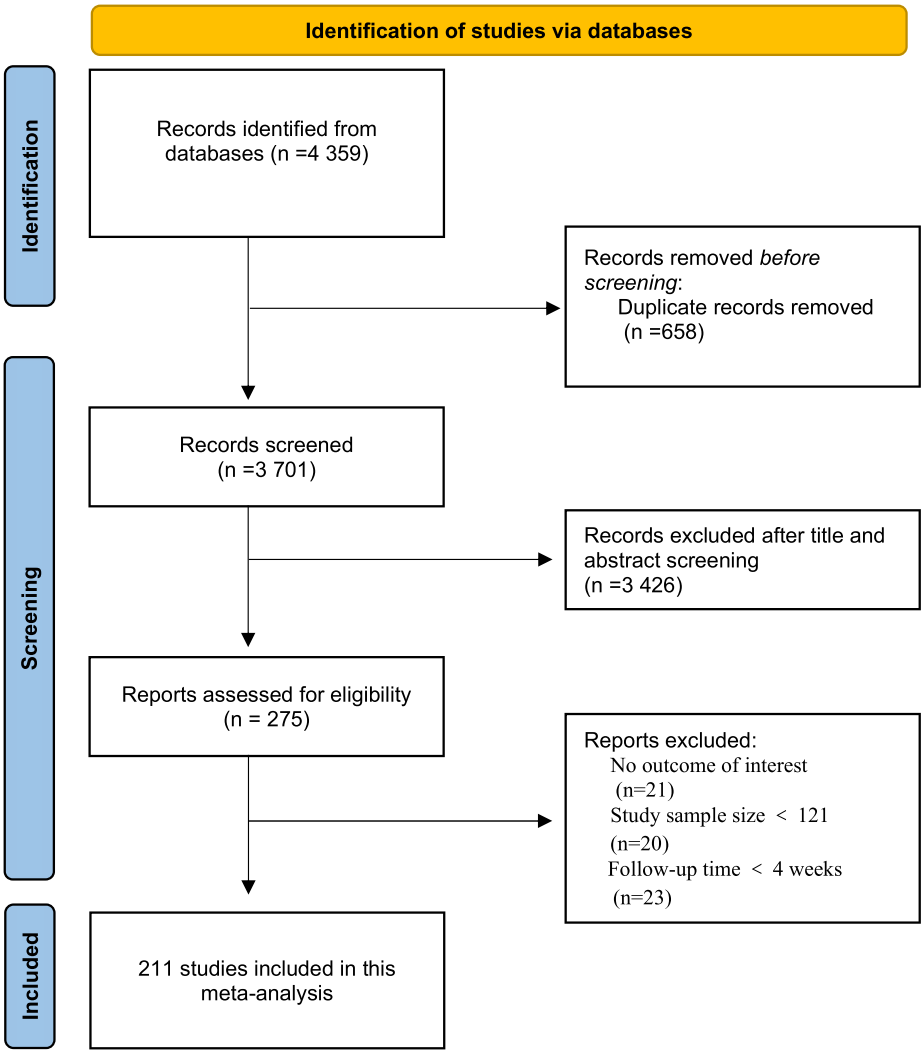


Fig. 1. Selection of studies for inclusion.

in the supplementary material. The studies were generally considered to be of good quality, with an average critical appraisal score of 8.3 out of 9 for all uncontrolled studies (Table S3). Among the controlled studies, 17 out of 28 studies had a NOS score of 7 or higher (Table S4). The included studies were conducted in various regions: Europe (n = 124), Asia (n = 44), North America (n = 26), South America (n = 5), Africa (n = 2), Oceania (n = 5), and across multiple continents (n = 5). This meta-analysis included a total of 13 368 074 participants, comprising 3 474 987 COVID-19 patients and 9 893 087 non-COVID-19 individuals. Non-COVID-19 individuals had a negative COVID-19 PCR, including healthy controls, symptomatic hospitalized patients, matched influenza patients, and matched individuals. Detailed information about the included 183 non-comparative studies and 28 comparative studies is shown in Tables S5 and S6, respectively. Participants' age ranged from 1 to 94 years, and the follow-up periods varied from 4 weeks to 24 months. Among the included studies, 40 had a follow-up period of at least 1 year. The analysis encompassed data from non-hospitalized (n = 86), hospitalized (n = 94), mixed non-hospitalized and hospitalized patients (n = 66), and intensive care unit (ICU) admissions (n = 8).

Meta-analyses of prevalence of persistent symptoms

The analysis included 211 studies that assessed the prevalence of persistent symptoms after COVID-19. The meta-analysis found significant heterogeneity in the prevalence of persistent symptoms across the studies, with the majority of symptoms exhibiting an $I^2 \geq 90\%$. Among the reported symptoms, fatigue (193 studies and 1 016 556 individuals), dyspnoea (175 studies and 1 001 000 individuals), post-traumatic stress disorder (PTSD) (27 studies and 10 319 individuals), anxiety (57 studies and 852 942 individuals), and depression (46 studies and 863 057 individuals) were the five most prevalent symptoms (Fig. 2, Table S7).

Subgroup analysis of prevalence of persistent symptoms

Patients aged ≤18 years had a lower prevalence of fatigue, hair loss, loss of smell, headache, cough, and loss of taste compared with those aged 19–≤60 years or >60 years. The subgroup analysis based on geographic location revealed that patients from Europe had a higher prevalence of fatigue, PTSD, anxiety, weight loss, and loss of smell compared with patients from Asia. Patients admitted

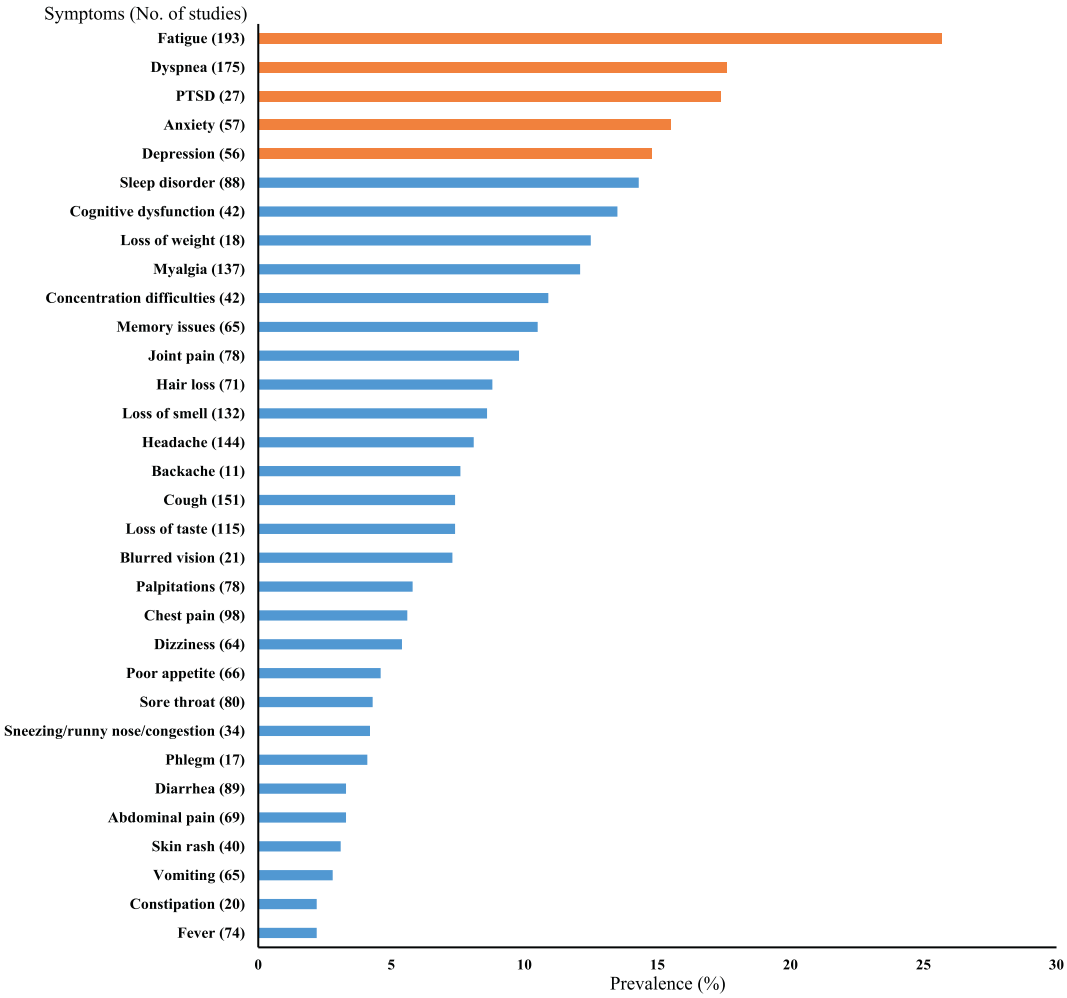


Fig. 2. Meta-analyses of prevalence of persistent symptoms.

to the ICU exhibited a higher prevalence of persistent symptoms, such as fatigue, dyspnoea, cough, and palpitations compared with non-hospitalized individuals. In comparison with non-hospitalized patients, hospitalized patients had a higher prevalence of hair loss and skin rash. The results of the subgroup analysis are presented in Table S8.

Meta-analyses of prevalence of persistent symptoms at different follow-up periods

The most frequently reported symptoms within a period of less than 3 months were fatigue, PTSD, anxiety, dyspnoea, and depression (Fig. 3(a), Table S9). Fatigue, PTSD, dyspnoea, myalgia, and backache were the most commonly reported symptoms between 3–6 months (Fig. 3(b), Table S10). The most commonly reported symptoms during the 6–12 months follow-up period were fatigue, dyspnoea, anxiety, depression, and cognitive dysfunction (Fig. 3(c), Table S11). The most commonly reported symptoms during the follow-up period of 12 months or more were fatigue, dyspnoea, sleep disorder, cognitive dysfunction, and joint pain (Fig. 3(d), Table S12).

Comparison of persistent symptom prevalence among COVID-19 and non-COVID-19 patients

A total of 28 studies provided adequate data for conducting a meta-analysis to compare the prevalence of persistent symptoms between the COVID-19 group and the comparator group. Meta-analyses were conducted for 25 symptoms that were reported by

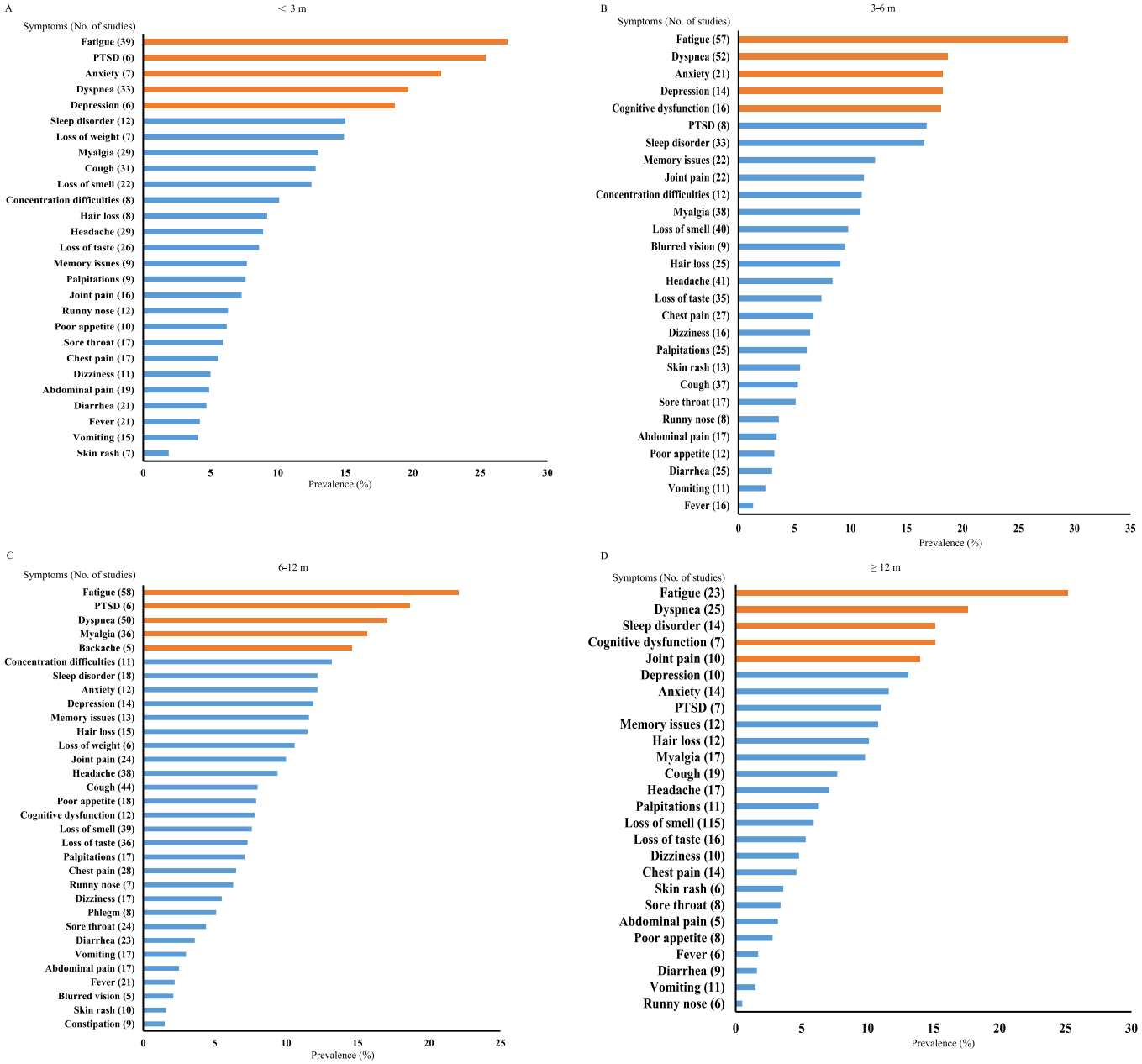


Fig. 3. Meta-analyses of prevalence of persistent symptoms at different follow-up periods. A: Meta-analyses of prevalence of persistent symptoms within a period of less than 3 months. B: Meta-analyses of prevalence of persistent symptoms between 3–6 months. C: Meta-analyses of prevalence of persistent symptoms between 6–12 months. D: Meta-analyses of prevalence of persistent symptoms during the follow-up period of 12 months or more.

more than five studies. Subgroup analyses based on follow-up periods were performed for these 25 symptoms.

The COVID-19 group exhibited a significantly higher prevalence of persistent respiratory symptoms, including dyspnoea, cough, and sore throat (Fig. S1). In the nervous system, COVID-19 group exhibited a significantly higher prevalence of memory issues, dizziness, headache, loss of taste, loss of smell, and concentration difficulties (Fig. S2). There were no significant differences in the prevalence of persistent abdominal pain, constipation, diarrhoea, and vomiting between the COVID-19 group and the comparator group (Fig. S3). The COVID-19 group exhibited a significantly higher prevalence of persistent joint pain and myalgia, (Fig. S4); palpitations and chest pain (Fig. S5); anxiety, sleep disorder, and depression (Fig. S6); skin rash, fatigue, and hair loss, but no significant difference in fever (Fig. S7).

Risk factors for persistent symptoms or long COVID

Among the 211 studies included in this study, 97 investigated risk factors associated with persistent symptoms or long COVID. The findings consistently identified several commonly reported risk factors, such as female gender, old age, disease severity during the acute phase of COVID-19, a high number of comorbidities, prolonged length of hospital stay, and high basic metabolic rate. Table S13 provides a summary of the results related to potential risk factors associated with persistent symptoms after COVID-19 or long COVID.

Discussion

In this systematic review and meta-analysis of 211 studies involving a total of 13 368 074 participants from various regions worldwide, we found that a significant proportion of patients experienced multiple clinically significant persistent symptoms affecting various organ systems. These symptoms continue to be highly prevalent even 1 year after the initial SARS-CoV-2 infection, supporting the widely reported belief that the recovery from COVID-19 can be a protracted process for certain individuals. The most commonly reported symptoms were fatigue, dyspnoea, PTSD, anxiety, depression, sleep disorder, cognitive dysfunction, weight loss, myalgia, and concentration difficulties, which align with the findings of previous studies and systematic reviews examining post-COVID symptoms. Subgroup analyses revealed that the prevalence of most persistent symptoms appeared to be higher among older patients, those with more severe illness in the acute phase of COVID-19, and individuals from Europe. Furthermore, the prevalence of most persistent symptoms in COVID-19 patients was significantly higher than that in individuals without COVID-19. Commonly reported risk factors associated with persistent symptoms after COVID-19 or long COVID included female gender, older age, disease severity in the acute phase of COVID-19, a high number of comorbidities, longer hospital stays, and high basic metabolic rate.

In this study, most of the meta-analyses exhibited high heterogeneity, likely because of variations in follow-up durations, data collection methods, and demographic characteristics among the included studies. The included studies had variable follow-up durations, ranging from 3 months to 2 years after the initial infection. In addition, the majority of the included studies assessed symptoms at multiple time points during follow-up. For most symptoms, the proportion of patients reporting them significantly decreased from the first 3 months to 12 months after infection. The prevalence of most symptoms exhibited a wave-like pattern in terms of decreasing trend over time, with only a few symptoms showing a gradual decrease. However, the prevalence of joint pain gradually increased over time. These findings are consistent with the results of previous large longitudinal studies, which demonstrated an overall decrease in the frequency of most symptoms over time, while a subset of symptoms showed an increased prevalence among a minority of individuals [18–20]. Previous studies have reported an association between increasing COVID-19 severity and a higher prevalence of persistent symptoms [21–23]. The results of our subgroup analysis revealed a significantly higher prevalence of persistent symptoms in ICU patients compared to non-hospitalized and hospitalized patients without ICU admission.

Previous studies have consistently shown significant variations in symptom prevalence among different age groups, with a higher prevalence observed in older adults and a lower prevalence observed in children [24–27]. We observed that children and adolescents tend to have a lower prevalence compared to adults, which is consistent with previous findings [28,29]. A previous meta-analysis [30] reported the prevalence of persistent symptoms after COVID-19 in children and adolescents, which is generally in line with our findings. The following hypotheses may help explain the lower prevalence of persistent symptoms in children and adolescents: First, children are believed to exhibit a stronger innate or mucosal immune response to COVID-19 when compared with adults, which can limit viral spread to the lower respiratory tract [31]. The naive nature of the paediatric adaptive immune system may contribute to milder acute-phase outcomes of COVID-19 [32,33]. Second, the utilization of angiotensin-converting enzyme 2 (ACE2) as the receptor for SARS-CoV-2 entry into human cells is supported by both bioinformatics modelling and *in vitro* experiments. The age-related variation in ACE2 expression, particularly the lower expression in children compared with adults, is an important biological feature that can significantly affect COVID-19 symptoms [34]. Third, the data reported by most included studies rely on participants' retrospective reporting of symptoms, which may introduce recall bias and potentially underestimate the prevalence. Last, children infected with SARS-CoV-2 are found to have significantly lower rates of positive PCR test results compared with adults, leading to a substantial number of cases being missed [35]. Consequently, the data reported by several included studies may not accurately reflect the true prevalence of persistent symptoms in children.

The reported prevalence of persistent symptoms in the majority of uncontrolled studies exhibits a wide range and relies on self-reports from patients, potentially inflating the prevalence estimates. Furthermore, these persistent symptoms are not exclusive to SARS-CoV-2; for instance, fatigue, muscle pain, headache, dyspnoea, and persistent dry cough may also manifest following influenza. Therefore, it is crucial to compare the incidence of these symptoms between individuals with COVID-19 and those without. In this study, we included controlled studies and conducted meta-analyses to compare the prevalence of these symptoms between the two populations. The findings revealed a significantly higher incidence of most symptoms in patients with COVID-19.

The aetiology of long COVID is believed to be multifactorial, with potentially overlapping factors [5,36–38]. Several hypotheses have been proposed to elucidate its pathogenesis, such as: (a) persistent SARS-CoV-2 reservoirs in tissues, (b) immune dysregulation, (c) alterations in the microbiota, (d) microvascular clotting accompanied by endothelial dysfunction, and (e) dysfunctional signalling in the brainstem or vagus nerve.

This study has several notable strengths. First, it represents the most up-to-date and comprehensive synthesis of global evidence, systematically analysing the prevalence of persistent symptoms after COVID-19 in the general population. Second, this study included a large number of patients across a wide age range, particularly children and adolescents. This enabled adequate subgroup analysis and increased statistical power for these age groups,

which were underrepresented in previous publications. Third, we included a substantial number of comparative studies and conducted meta-analyses to compare the prevalence of various persistent symptoms between COVID-19 patients and individuals without COVID-19. Fourth, the inclusion of studies with various follow-up periods enabled us to investigate the temporal dynamics and evolution of individual symptoms over time. In addition, we included 37 studies comprising 2 689 773 individuals with a follow-up period of 12 months or more. The results of long-term follow-up on a large sample of individuals aid in enhancing our understanding of long COVID.

However, it is important to acknowledge several limitations inherent to both the present study and the studies included. First, the symptom data reported in most studies included were obtained using various self-report tools, likely because of the limited availability of comprehensive symptom burden assessment tools during the initial stages of the COVID-19 pandemic. Differences in measurement tools may contribute to high heterogeneity among the studies. Moreover, the definitive attribution of self-reported persistent symptoms to COVID-19 infection could not be confirmed. To partially address this limitation, we compared the prevalence of persistent symptoms between COVID-19 patients and individuals without COVID-19. However, future studies should use standardized measurement tools to prevent or at least minimize the possibility of erroneously attributing symptoms because of another disease to long- COVID. Second, we could not summarize the overall prevalence of long COVID based on the studies included. This is because of variations in the number of reported symptoms and the definitions of long COVID across different studies. To evaluate the prevalence of long COVID more accurately, we assessed the prevalence of each specific symptom. Third, we did not examine the effect of vaccination because of the limited number of studies reporting vaccination status. With the emergence of new viral variants and the increasing number of vaccinated individuals, future studies will yield more data on whether similar symptom patterns are observed with these new variants and whether the vaccines provide protection against them. Another limitation of this study is the geographical homogeneity observed among the included studies, with the majority of data originating from European regions. To obtain a comprehensive understanding of the burden and long-term effects of COVID-19, it is crucial to conduct additional research in regions characterized by a high prevalence of low-income and middle-income countries, such as Africa. These investigations will facilitate the proper allocation and utilization of resources, and the development of culturally appropriate intervention strategies.

This meta-analysis provides a thorough review of the prevalence of and risk factors for persistent symptoms following COVID-19. The findings underscore the importance of long-term monitoring and support for individuals recovering from COVID-19, especially among older adults, individuals with severe illness, and those with multiple comorbidities. Additional research is necessary to advance our understanding of the underlying mechanisms and risk factors related to long COVID, facilitating the development of targeted interventions and health care strategies for this population.

Author contributions

DL designed the study. DL, PW, and XJL performed the searches. PW, XJL, and GW conducted the data extraction. PW and XJL assessed the risk of bias. XPC and FK conducted the statistical analysis. DL and BJM drafted and edited the manuscript. BL and SS provided oversight, critical evaluation, and verification of the manuscript. All authors approved the final version of the manuscript. DL, BJM, PW, and XJL contributed equally to this work.

Transparency declaration

No new data were generated during the present study. All data used for meta-analyses in the present study are from published articles. Data are available from these published articles or the original study authors.

Authors declare that they have no conflicts of interest.

No external funding was received for this study.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cmi.2023.10.016.

References

[1] Nalbandian A, Desai AD, Wan EY. Post-COVID-19 condition. Annu Rev Med 2023;74:55–64. https://doi.org/10.1146/annurev-med-043021-030635.
[2] Soriano JB, Murthy S, Marshall JC, Relan P, Diaz JV. WHO clinical case definition working group on post-COVID-19 condition. A clinical case definition of post-COVID-19 condition by a Delphi consensus. Lancet Infect Dis 2022;22:e102–7. https://doi.org/10.1016/S1473-3099(21)00703-9.
[3] Venkatesan P. NICE guideline on long COVID. Lancet Respir Med 2021;9:129. https://doi.org/10.1016/S2213-2600(21)00031-X.
[4] Sivan M, Taylor S. NICE guideline on long covid. BMJ 2020;371:m4938. https://doi.org/10.1136/bmj.m4938.
[5] Davis HE, McCorkell L, Vogel JM, Topol EJ. Long COVID: major findings, mechanisms and recommendations. Nat Rev Microbiol 2023;21:133–46. https://doi.org/10.1038/s41579-022-00846-2.
[6] Alkodaymi MS, Omrani OA, Fawzy NA, Shaar BA, Almamlouk R, Riaz M, et al. Prevalence of post-acute COVID-19 syndrome symptoms at different follow-up periods: a systematic review and meta-analysis. Clin Microbiol Infect 2022;28:657–66. https://doi.org/10.1016/j.cmi.2022.01.014.
[7] Behnood SA, Shafran R, Bennett SD, Zhang AXD, O'Mahoney LL, Stephenson TJ, et al. Persistent symptoms following SARS-CoV-2 infection amongst children and young people: a meta-analysis of controlled and uncontrolled studies. J Infect 2022;84:158–70. https://doi.org/10.1016/j.jinf.2021.11.011.
[8] Chen C, Haupert SR, Zimmermann L, Shi X, Fritsche LG, Mukherjee B. Global prevalence of post-coronavirus disease 2019 (COVID-19) condition or long COVID: a meta-analysis and systematic review. J Infect Dis 2022;226:1593–7. https://doi.org/10.1093/infdis/jiac136.
[9] Ceban F, Ling S, Lui LMW, Lee Y, Gill H, Teopiz KM, et al. Fatigue and cognitive impairment in post-COVID-19 syndrome: a systematic review and meta-analysis. Brain Behav Immun 2022;101:93–135. https://doi.org/10.1016/j.bbi.2021.12.020.
[10] Booth A, Clarke M, Ghersi D, Moher D, Petticrew M, Stewart L. An international registry of systematic-review protocols. Lancet 2011;377:108–9. https://doi.org/10.1016/S0140-6736(10)60903-8.
[11] Munn Z, Moola S, Lisy K, Riitano D, Tufanaru C. Methodological guidance for systematic reviews of observational epidemiological studies reporting prevalence and cumulative incidence data. Int J Evid Based Healthc 2015;13:147–53. https://doi.org/10.1097/XEB.0000000000000054.
[12] Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. Eur J Epidemiol 2010;25:603–5. https://doi.org/10.1007/s10654-010-9491-z.
[13] Cumpston M, Li T, Page MJ, Chandler J, Welch VA, Higgins JP, et al. Updated guidance for trusted systematic reviews: a new edition of the Cochrane Handbook for Systematic Reviews of Interventions. Cochrane Database Syst Rev 2019;10:ED000142. https://doi.org/10.1002/14651858.ED000142.
[14] Altman DG, Egger M, Smith GD. Systematic reviews in health care: meta-analysis in context. 2nd ed. UK: BMJ Publishing Group; 2001.
[15] Alfirevic Z, Stampalija T, Gyte GM. Fetal and umbilical Doppler ultrasound in normal pregnancy. Cochrane Database Syst Rev 2010;8:CD001450. https://doi.org/10.1002/14651858.CD001450.pub3.
[16] Wei S, Wo BL, Qi HP, Xu H, Luo ZC, Roy C, et al. Early amniotomy and early oxytocin for prevention of, or therapy for, delay in first stage spontaneous labour compared with routine care. Cochrane Database Syst Rev 2013:CD006794. https://doi.org/10.1002/14651858.CD006794.pub4.
[17] Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al., editors. Cochrane handbook for systematic reviews of interventions version 6. 4 Cochrane; 2023.
[18] Huang L, Yao Q, Gu X, Wang Q, Ren L, Wang Y, et al. 1-year outcomes in hospital survivors with COVID-19: a longitudinal cohort study. Lancet 2021;398:747–58. https://doi.org/10.1016/S0140-6736(21)01755-4.
[19] Petersen MS, Kristiansen MF, Hanusson KD, Danielsen ME, ÁS B, Gaini S, et al. Long COVID in the Faroe Islands: a longitudinal study among nonhospitalized patients. Clin Infect Dis 2021;73:e4058–63. https://doi.org/10.1093/cid/ciaa1792.
[20] Petersen MS, Kristiansen MF, Hanusson KD, Foldbo BM, Danielsen ME, ÁS B, et al. Prevalence of long COVID in a national cohort: longitudinal measures

from disease onset until 8 months' follow-up. Int J Infect Dis 2022;122:437–41. https://doi.org/10.1016/j.ijid.2022.06.031.
[21] Ahmed GK, Khedr EM, Hamad DA, Meshref TS, Hashem MM, Aly MM. Long term impact of Covid-19 infection on sleep and mental health: a cross-sectional study. Psychiatry Res 2021;305:114243. https://doi.org/10.1016/j.psychres.2021.114243.
[22] Evans RA, McAuley H, Harrison EM, Shikotra A, Singapuri A, Sereno M, et al. Physical, cognitive, and mental health impacts of COVID-19 after hospitalisation (PHOSP-COVID): a UK multicentre, prospective cohort study. Lancet Respir Med 2021;9:1275–87. https://doi.org/10.1016/S2213-2600(21)00383-0.
[23] Fang X, Ming C, Cen Y, Lin H, Zhan K, Yang S, et al. Post-sequelae one year after hospital discharge among older COVID-19 patients: a multi-center prospective cohort study. J Infect 2022;84:179–86. https://doi.org/10.1016/j.jinf.2021.12.005.
[24] Abdelrahman MM, Abd-Elrahman NM, Bakheet TM. Persistence of symptoms after improvement of acute COVID19 infection, a longitudinal study. J Med Virol 2021;93:5942–6. https://doi.org/10.1002/jmv.27156.
[25] Martinez AE, Banderet F, Labhardt ND, Battegay M. Long-term outcome after SARS-CoV-2 infection in healthcare workers: a single centre cohort study. Swiss Med Wkly 2021;151:w30094. https://doi.org/10.4414/smw.2021.w30094.
[26] Yomogida K, Zhu S, Rubino F, Figueroa W, Balanji N, Holman E. Post-acute sequelae of SARS-CoV-2 infection among adults aged ≥18 years—long Beach, California, April 1-December 10, 2020. MMWR Morb Mortal Wkly Rep 2021;70:1274–7. https://doi.org/10.15585/mmwr.mm7037a2.
[27] Mohiuddin Chowdhury ATM, Karim MR, Ali MA, Islam J, Li Y, He S. Clinical characteristics and the long-term post-recovery manifestations of the COVID-19 patients-a prospective multicenter cross-sectional study. Front Med (Lausanne) 2021;8:663670. https://doi.org/10.3389/fmed.2021.663670.
[28] Zimmermann P, Curtis N. Coronavirus infections in children including COVID-19: an overview of the epidemiology, clinical features, diagnosis, treatment and prevention options in children. Pediatr Infect Dis J 2020;39:355–68. https://doi.org/10.1097/INF.0000000000002660.
[29] Zimmermann P, Curtis N. Why is COVID-19 less severe in children? A review of the proposed mechanisms underlying the age-related difference in severity of SARS-CoV-2 infections. Arch Dis Child 2020;106:429–39. https://doi.org/10.1136/archdischild-2020-320338.
[30] Lopez-Leon S, Wegman-Ostrosky T, Ayuzo Del Valle NC, Perelman C, Sepulveda R, Rebolledo PA, et al. Long-COVID in children and adolescents: a systematic review and meta-analyses. Sci Rep 2022;12:9950. https://doi.org/10.1038/s41598-022-13495-5.
[31] Yoshida M, Worlock KB, Huang N, Lindeboom RGH, Butler CR, Kumasaka N, et al. Local and systemic responses to SARS-CoV-2 infection in children and adults. Nature 2022;602:321–7. https://doi.org/10.1038/s41586-021-04345-x.
[32] Cohen CA, Li APY, Hachim A, Hui DSC, Kwan MYW, Tsang OTY, et al. SARS-CoV-2 specific T cell responses are lower in children and increase with age and time after infection. Nat Commun 2021;12:4678. https://doi.org/10.1038/s41467-021-24938-4.
[33] Pierce CA, Preston-Hurlburt P, Dai Y, Aschner CB, Cheshenko N, Galen B, et al. Immune responses to SARS-CoV-2 infection in hospitalized pediatric and adult patients. Sci Transl Med 2020;12. https://doi.org/10.1126/scitranslmed.abd5487.
[34] Bunyavanich S, Do A, Vicencio A. Nasal gene expression of angiotensin-converting enzyme 2 in children and adults. JAMA 2020;323:2427–9. https://doi.org/10.1001/jama.2020.8707.
[35] Langeland N, Cox RJ. Are low SARS-CoV-2 viral loads in infected children missed by RT-PCR testing? Lancet Reg Health Eur 2021;5:100138. https://doi.org/10.1016/j.lanepe.2021.100138.
[36] Crook H, Raza S, Nowell J, Young M, Edison P. Long covid-mechanisms, risk factors, and management. BMJ 2021;374:n1648. https://doi.org/10.1136/bmj.n1648.
[37] Oronsky B, Larson C, Hammond TC, Oronsky A, Kesari S, Lybeck M, et al. A review of persistent post-COVID syndrome (PPCS). Clin Rev Allergy Immunol 2023;64:66–74. https://doi.org/10.1007/s12016-021-08848-3.
[38] Chen B, Julg B, Mohandas S, Bradfute SB. Viral persistence, reactivation, and mechanisms of long COVID. eLife 2023;12:e86015. https://doi.org/10.7554/eLife.86015.